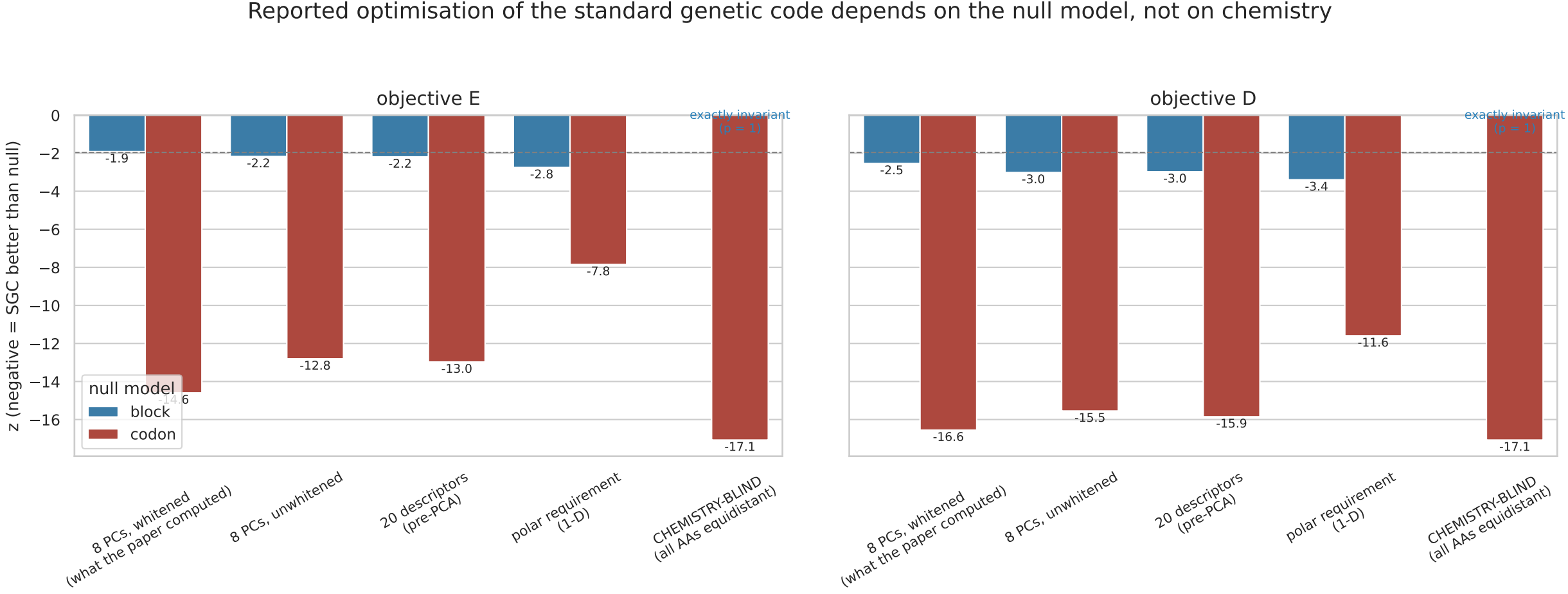
Reported optimisation of the standard genetic code depends on the null model, not on chemistry
objective E
objective D
z (negative = SGC better than null)
0
−2
−4
−6
−8
−10
−12
−14
−16
null model
block
codon
-1.9
-14.6
-2.2
-12.8
-2.2
-13.0
-2.8
-7.8
exactly invariant
(p = 1)
-17.1
-2.5
-16.6
-3.0
-15.5
-3.0
-15.9
-3.4
-11.6
exactly invariant
(p = 1)
-17.1
8 PCs, whitened
(what the paper computed)
8 PCs, unwhitened
20 descriptors
(pre-PCA)
polar requirement
(1-D)
CHEMISTRY-BLIND
(all AAs equidistant)